

Original Article

Tongqiao Huoxue Decoction inhibits ferroptosis by facilitating ACSL4 ubiquitination degradation for neuroprotection against cerebral ischemia-reperfusion injury

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ABSTRACT

Background: Cerebral ischemia-reperfusion injury (CIRI) refers to brain tissue injury caused by the temporary interruption of cerebral blood flow ischemia followed by the restoration of reperfusion, which is the main cause of post-stroke brain injury. A traditional Chinese herbal preparation called Tongqiao Huoxue Decoction (TQHX) has shown promise in reducing CIRI in rats. However, the mechanism of this herbal preparation for CIRI remains unclear.

Purpose: This study aimed to evaluate the therapeutic effect of TQHX extract on rats with CIRI and to further explore the underlying mechanisms.

Methods: The active ingredients of TQHX extract were quantified by the high-performance liquid chromatography (HPLC) condition. We conducted thorough investigations to assess the effects of TQHX on CIRI and ferroptosis using oxygen-glucose deprivation/reperfusion (OGD/R)-treated PC12 cells as an in vitro model and transient middle cerebral artery occlusion (tMCAO) animals as an in vivo model. The neurological score assessment was performed to evaluate the neuroprotective effects of TQHX extract on tMCAO rats. Using histologic methods to study the extent of cerebral infarction, blood-brain barrier, and rat brain tissue. We examined the impact of TQHX on ferroptosis-related markers of Fe²⁺, superoxide dismutase (SOD), reactive oxygen species (ROS), and malondialdehyde (MDA) in the brain tissue. In addition, the expression of key proteins and markers of ferroptosis, as well as key factors associated with Acyl-CoA synthetase long-chain family member 4 (ACSL4) were detected by Western blot and quantitative real-time PCR (RT-qPCR).

Results: TQHX extract could decrease the Longa score and extent of cerebral infarction of tMCAO rats, which exerted the function of neuroprotection. Additionally, TQHX treatment efficiently decreased levels of MDA and ROS while increasing the expression of SOD and ferroptosis-related proteins including ferritin heavy chain 1 (FTH1) and glutathione peroxidase 4 (GPX4) at the transcription and translation level. Meanwhile, TQHX provided strong protection against oxidative stress and ferritin accumulation by increasing the ubiquitination and degradation of ACSL4. The injection of OE-ACSL4 reversed the effects of TQHX on neuroprotection and ferroptosis inhibition in PC12 cells. The injection of shACSL4 reversely validate the crucial role of ACSL4 in CIRI rat treatment.

Abbreviations: ACSL4, Acyl-CoA Synthetase Long Chain Family Member 4; AAs, Arachidonic acids; BSA, bovine serum albumin; CHX, Cycloheximide; CIRI, Cerebral ischemia/reperfusion injury; Co-IP, Co-Immunoprecipitation; ECA, external carotid artery; ELISA, Enzyme-linked immunosorbent assays; GPX4, glutathione peroxidase 4; HE, Hematoxylin and Eosin Staining; HPLC, High-Performance Liquid Chromatography; I/R, ischemia/reperfusion; ICA, internal carotid artery; MCA, middle cerebral artery; MDA, malondialdehyde; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide; OGD/R, oxygen-glucose deprivation/reperfusion; PBS, phosphate-buffered saline; PUFAs, polyunsaturated fatty acids; RT-qPCR, quantitative real-time PCR; ROS, reactive oxygen species; SOD, superoxide dismutase; TCM, Traditional Chinese Medicine; TQHX, Tongqiao Huoxue Decoction; tMCAO, transient middle cerebral artery occlusion; TTC, Triphenyltetrazolium Chloride Staining; WB, Western Blot assay.

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Conclusion: This work shows that TQHX promotes the ubiquitination-mediated degradation of ACSL4, which improves oxidative stress and inhibits the beginning of ferroptosis in cells. TQHX provides a possible path for additional research in CIRI therapies, advancing translational investigations.

Introduction

Stroke is currently the third leading cause of death and severe permanent impairment worldwide. In recent years, there has been an increase in the overall number of stroke incidents, prevalence of strokes, stroke-related disability, and stroke-related death (2021). Approximately 80 % of all stroke cases are ischemia stroke due to intracerebral artery embolism, which has emerged as one of the world's major causes of mortality and permanent disability (Liao et al., 2019; Wu et al., 2018). Currently, t-PA thrombolytic therapy and surgical thrombectomy are the two major therapies for ischemic stroke. Recanalization of the cerebral arteries is the goal of these therapies, but they might paradoxically cause cerebral ischemia-reperfusion injury (CIRI), worsening the extent of brain damage, including oxidative stress, neuroinflammation, mitochondrial dysfunction, and programmed cell death (Zhao et al., 2023a). Thus, to prevent brain damage before and during recanalization, increase the therapeutic window for intervention, and enhance functional outcomes, it is imperative to develop therapeutic drugs for neuroprotection in acute ischemic stroke (Paul and Candelario-Jalil, 2021).

Traditional Chinese Medicine (TCM) has recently shown particular benefits in the treatment of CIRI (Yang et al., 2021). Numerous scientists have conducted relevant studies, demonstrating that TCM exhibits neuroprotective properties against CIRI, encompassing antioxidant and anti-inflammatory effects. Unfortunately, there remains a paucity of research elucidating the underlying mechanisms through which TCM exerts its therapeutic effects on CIRI, thereby presenting substantial challenges in the development of treatments for this condition.

Tongqiao Huoxue decoction (TQHX) is a type of decoction that is used to promote blood circulation and remove blood stasis. It was first described in Wang Qingren's *Medical Forest Reform* during the Qing Dynasty (Zhou et al., 2022), mainly composed of eight herbs, such as *Carthamus tinctorius*, *Zingiber officinale*, *Ziziphus jujuba*. Recent research has demonstrated that TQHX can considerably lessen neurological damage during cerebral ischemia-reperfusion in rats, whose therapeutic effectiveness is manifested through various neuroprotective pathways (Yuan et al., 2022b). Notably, TQHX is primarily known for its ability to alleviate oxidative stress (Wang et al., 2012), a crucial factor in neuronal damage resulting from CIRI. By removing free radicals and enhancing antioxidant defenses, TQHX mitigates oxidative damage to neuronal cells, protects neurons from CIRI, and prevents apoptosis. Previous studies have demonstrated that TQHX can safeguard neurons against CIRI injury and prevent apoptosis by regulating the pathway linked to ferroptosis (Yuan et al., 2022b). Therefore, TQHX may be an effective treatment for CIRI by reducing ferroptosis. However, the impact of TQHX on CIRI-related ferroptosis and its underlying mechanisms is still unclear.

Ferroptosis is a form of regulated cell death that is characterized by iron-dependent lipid peroxidation (Li et al., 2020a). It is distinct from apoptosis and necrosis, as it involves the accumulation of lipid reactive oxygen species (ROS) (Zhang et al., 2023b). During this process, lipid hydroperoxides fatally accumulate within cell membranes, causing damage to the membranes and ultimately leading to cell death. In CIRI, cerebral ischemia leads to cellular hypoxia (Zhang et al., 2022b). Reperfusion can aggravate oxidative stress by increasing intracellular Fe^{2+} and causing iron-dependent oxidative reactions that produce large amounts of oxygen radicals and exacerbate lipid peroxidation of cell membranes, ultimately leading to ferroptosis (Tang and Kang, 2023). Acyl-CoA synthetase long-chain family member 4 (ACSL4) is a crucial regulator of ferroptosis, particularly arachidonic acid, into phospholipids. This enhances lipid peroxidation, which is a crucial step in

ferroptosis (Tuo et al., 2022). Conversely, reducing or inhibiting ACSL4 has been shown to protect cells from ferroptosis by decreasing the amount of substrates required for lipid peroxidation (Zhang et al., 2022a). Nevertheless, whether TQHX modulates the expression of ACSL4 thus regulating ferroptosis in CIRI remains unclear.

This study aimed at investigating the potential mechanisms of TQHX in treating CIRI. The transient middle cerebral artery occlusion (tMCAO) model was structured in vivo, and the OGD/R cell model was structured in vitro to verify the regulatory effects of TQHX on ferroptosis to alleviate CIRI.

Materials and methods

Drugs and reagents

The source of *Paeoniae Radix Rubra* (Red Paeony Root), *Chuanxiong Rhizoma* (Persimmon), *Persicae Semen* (Peach Seed), *Jujubae Fructus* (Jujube), *Carthamus tinctorius* (Red Flower), *Allium fistulosum* (Scallion), *Zingiber officinale* (Fresh Ginger) and *Moschus* (Musk) shown in Table S1 was purchased from Anhui Tonghuatang Chinese Medicine Drinking Tablets Technology Co (Anhui, China).

Edaravone (P816062), the positive control for this study, was acquired from Shanghai Macklin Biochemical Co., Ltd. located in Shanghai, China. Anti-FTH1 (ab183781, dilution 1:1000), anti-GPX4 (ab125066, dilution 1:2000), and anti-ACSL4 (ab155282, dilution 1:1000) were purchased from Abcam (USA). Ubiquitin (P4D1) Mouse Antibody (#3936, dilution 1:1000) and GAPDH (D16H11) XP® Rabbit Antibody (#5174, dilution 1:5000) were acquired from Cell Signaling Technology. The Iron Assay Kit (Colorimetric) (ab83366), anti-FACL4 antibody (ab227256, 1:200), and Goat Anti-Rabbit IgG H&L (Alexa Fluor® 594) preadsorbed (ab150084, 1:1000) were all purchased from Abcam (USA). The *In Vitro* ROS/RNS Assay (STA-347) was purchased from Cell Biolabs, Inc. Additionally, the Total Superoxide Dismutase Assay Kit with WST-8 (S0101S) and Lipid Peroxidation MDA Assay Kit (S0131S) were both purchased from Beyotime Biotechnology.

Animals

Sibeifu Biotechnology Co. supplied male 12-week-old Sprague-Dawley rats weighing between 250 and 300 gs. The rats' house was set to a controlled 12-hour light/dark cycle with unlimited access to food and water. The Institutional Animal Care and Use Committee of the Nanjing University of Chinese Medicine set regulations that were followed in all animal studies (Approval No. 202307A141). For one week before the investigations began, the rats were housed in a controlled environment with a temperature of 24.0 ± 2.0 °C and a relative humidity of 55.0 ± 5.0 %.

tMCAO model establishment

The middle cerebral artery (MCA) origin was blocked by inserting a nylon filament into the carotid artery, creating the tMCAO model. In brief, during the surgical operation, rats were sedated with 3 % phenobarbital sodium. Then an 0.38 mm diameter silicone-coated nylon monofilament was injected into the internal carotid artery (ICA) through the external carotid artery (ECA), growing till it obstructed the MCA's original source. When the brain was occluded, we tracked cerebral blood flow using a laser speckle contrast imager. Throughout the process, vital signs such as blood pressure, oxygen saturation, and heart rate, were continuously checked and the body temperature was kept at 37 ± 0.5 °C.

After 120 min of ischemia, the filament was removed to start reperfusion (Lee et al., 2014). The sham-operated rats performed the same process with the exception of inserting the filament into the ICA (Li et al., 2020b).

Group assignment and treatment

Animal welfare and safety are given priority in treatment methods and group assignments, which are guided by recognized ethical principles and procedures. The Sham, tMCAO, tMCAO + TQHX low dose (TQHX-L), tMCAO + TQHX medium dose (TQHX-M), tMCAO + TQHX high dose (TQHX-H), and tMCAO + Edaravone (positive control) were the six groups to which the rats were randomly assigned. The number of animals used in each experiment was decided via power analysis to minimize animal suffering and provide adequate statistical power. Two hours after reperfusion, the TQHX treatment groups received intraperitoneal injections of TQHX-L (3 g/kg), TQHX-M (6 g/kg), and TQHX-H (12 g/kg) twice a day (Yuan et al., 2022a). Additionally, twice daily intraperitoneal injections of 10 mg/kg of Edaravone, the positive control, were administered (Ha et al., 2020). The same approach was followed while giving the same amount of saline solution to the rats in the Sham group and tMCAO group. This formula was used to determine the dosage: given dosage multiplied by the animal equivalent dosage coefficient (based on body surface area) equals the clinically usual dosage.

High performance liquid chromatography condition (HPLC)

Using model 2695 with a YMC—ODS column (250 × 4.6 mm, 5 μm) and a gradient elution method at 1 ml/min, the samples were separated by HPLC system (Waters, USA). Solvent A (water with 0.1 % formic acid) and solvent B (acetonitrile) made up the mobile phase, which allowed for effective separation. The run lasted for 110 min, and the last cleaning and equilibration required 10 μl of injection volume. The temperature inside the column was kept at 30°C. Gradient elution was performed (flow rate of 1.0 ml/min; 0–10 min, 99 % A; 10–13 min, 99 %→98 % A; 13–63 min, 98 %→95 % A; 63–110 min, 85 %→65 % A). Primary compounds extracted from TQHX were analyzed and identified.

Neurological score

Rats with CIRI have their neurological function evaluated using the Longa score method. The neurological function was measured on a scale of 0–4, where 0 represents normal function and 4 represents the most severe defect. After a 24-hour reperfusion interval and a 120-minute ischemia period, assessments were performed. Based on the following standards, the neurological condition of the rats was evaluated: A score of zero indicates the absence of a neurological abnormality. An anterior claw on the paralyzed side cannot be fully extended, as indicated by a score of 1. When moving, a score of two indicates a tendency to turn in the direction of the paralyzed side. A score of three indicates a propensity to sway toward the paralyzed side when moving. A score of 4 indicates a level of unconsciousness that prevents the subject from being able to move on their own. The study found a positive correlation between the degree of motor dysfunction and neurological impairment (Liu et al., 2023).

Triphenyl tetrazolium chloride (TTC) staining

Following the rat's execution by guillotine, the brain was extracted within ten minutes and put in a phosphate-buffered saline (PBS) solution at 0–4°C. It was then frozen for thirty minutes at −20°C. Brain slices were then sliced to a thickness of 2 mm and put in a 2 % red tetrazolium solution. The mixture was kept in a water bath at 37°C and shielded from light for thirty minutes. The containers were carefully shaken every five minutes to guarantee complete staining. After rinsing the slices in a PBS solution for three to five minutes, they were immediately photographed

and fixed for six hours in 10 % neutral formaldehyde. For image analysis using the Pathology Graphic Analysis System, the lowest side of each slice was chosen. The total area and infarct area of each slice were measured, and the infarct volume of each layer was computed by multiplying its thickness by the layer's infarct area. The total infarct volume was the sum of the infarct volumes in each layer.

Evans blue extravasation assay

Following CIRI in rats, the blood-brain barrier integrity was assessed employing the Evans blue extravasation assay, which included injecting a 2 % Evans Blue dye solution intravenously, circulating the blood for one to two hours, perfusing the heart with ice-cold saline, obtaining brain tissue, homogenizing different brain regions, and extracting the dye using trichloroacetic acid. Spectrophotometry was used to determine the quantity of Evans Blue dye at 620 nm about the weight of the removed brain regions. Disruption of the blood-brain barrier was assessed by contrasting the results between groups.

Nissl staining

After being preserved in 4 % paraformaldehyde, the brain tissue was embedded in paraffin. Glass slides were used to hold the 5 μm thick slices. Following deparaffinization, the slices underwent rehydration, 10 min of cresyl violet staining, 95 % ethanol differentiation, dehydration, clearing, and mounting. Counting both necrotic and normal neurons within the ischemic penumbra was done by a blinded researcher. Neuronal survival rate (%) = (number of normal neurons / total number of neurons) × 100.

Intracellular ROS detection

Following CIRI, intracellular ROS in brain tissue and PC12 cells were detected using fluorescent probes, carboxy-H2DCFDA (Sigma-Aldrich), and DHE (Sigma-Aldrich). Brain tissues were homogenized in PBS before the supernatant was centrifuged to carry out the DHE staining process. After that, 10 μm DHE was incubated for 30 min at 37°C with the supernatant. In order to identify ROS in cultured cells, the cells were seeded onto 6-well plates and subjected to a 30-minute 37°C DCF (10 μm) treatment. Using a SpectraMax microplate reader (Molecular Devices, San Jose, CA) at 488 nm excitation and 525 nm emission wavelengths, fluorescence was detected.

Cell culture and treatment

We assessed the effect of TQHX on ferroptosis using PC12 cells produced by oxygen-glucose deprivation/reoxygenation as an in vitro model of CIRI. Cycloheximide (CHX) inhibits the transcription and translation processes involved in protein synthesis, preventing cells from synthesizing new proteins, which was added aiming at exploring the mechanism of ACSL4 degradation. The Model + CHX group and Model + CHX + TQHX (12 g/kg) group were the two groups to which the cells were assigned. The PC12 cells were taken from the Chinese Academy of Sciences Cell Bank in Shanghai, China. The cells were then cultured in Thermo Fisher Scientific, Inc.'s Gibco's Modified Eagle's medium (DMEM), which was enhanced with 10 % fetal bovine serum (FBS), 100 U/ml penicillin, and 100 μg/ml streptomycin. The penicillin and streptomycin came from Cytiva's HyClone. The cells were grown in a humidified atmosphere with 5 % CO₂ at 37°C. In order to eliminate oxygen, PC12 cells were pre-incubated for two hours in a hypoxic chamber before being cultivated in glucose-free, serum-free DMEM media for the oxygen-glucose deprivation/reperfusion (OGD/R) model. After that, the cells were exposed to hypoxic conditions for a full day at 37°C (5 % CO₂, 1 % O₂, and 94 % N₂). The cells were then taken out of the hypoxic chamber and cultured in a DMEM medium containing high glucose. Before being exposed to OGD/R, cells were treated to several

concentrations of TQHX (10, 20, or 40 μM) for two hours to evaluate its protective qualities. The animals received TQHX by gavage for a consecutive seven days. To ascertain the dosage per animal and the required amount during the gavage cycle, blood was extracted 1–2 h after the last gavage. The rats were put to sleep and then given 8–10 cc of TQHX-containing blood. After 30 min at room temperature, the supernatant of centrifuged blood containing TQHX is collected (not higher than 20°C). After that, the supernatant's constituent parts were looked at. Additionally, to examine the impact of TQHX on Fe^{2+} levels, 25 μM FC was added to the culture medium.

3- (4,5-Dimethylthiazol-2-yl)–2,5-diphenyltetrazolium bromide (MTT) assay

Adjust the concentration of cells to the log phase and add 100 μl per well on a plate to adjust the density of cells to 1000–10,000 wells. Fill the edge of the wells with sterile PBS. Finally, observe the cells using an inverted microscope. Incubate the PC12 cells at 5 % CO_2 , 37°C until they form a monolayer over the bottom of the 96-well flat-bottomed plate. Then, add a concentration gradient of the drug and incubate the cells at 5 % CO_2 , 37°C for 16–48 h. Fill each well with 20 μl of MTT solution (5 mg/ml, or 0.5 % MTT), and then let it sit for 4 h. Centrifuge the medication and throw away the culture solution if it reacts with MTT. After that, give the wells a thorough two to three PBS rinses before adding the MTT-containing culture solution. Aspirate the culture medium out of the wells and stop the incubation. To fully dissolve the crystals, add 150 μl of dimethyl sulfoxide to each well and shake the mixture at a low pace for ten minutes. Calculate each well's absorbance using the enzyme immunoassay detector at OD490 nm. Prepare the control wells (containing cells, drug dissolution medium at the same concentration, culture medium, MTT, and dimethyl sulfoxide) and zero wells (containing culture medium, MTT, and dimethyl sulfoxide) simultaneously.

Oxidative stress and Fe^{2+} content assessment

Enzyme-linked immunosorbent assays (ELISA), superoxide dismutase (SOD) assays, and colorimetric assays were used by the manufacturer's instructions to measure the levels of malondialdehyde (MDA), as well as to ascertain oxidative stress and Fe^{2+} levels in brain tissue or PC12 cells after CIRC. The samples for testing were added to the solid phase carrier after anti-MDA and SOD antibodies were applied to its surface. The SOD and MDA in the samples interacted with the antibodies on the surface of the solid phase carrier. The solid-phase carrier was cleaned, and then the antibody tagged with an enzyme was added and attached to the reaction. The amount was then quantitatively assessed based on the color intensity when the substrate for the enzyme reaction was added. The o-phenanthroline method, which depended on the color shift of the complex produced by Fe^{2+} and o-phenanthroline, was used to determine the Fe^{2+} content. Each cuvette was pipetted with a specific amount of the Fe^{2+} standard solution—0.01 mg per milliliter. Subsequently, the hydroxylamine hydrochloride, o-phenanthroline solution, and sodium acetate buffer solution were sequentially added to every cuvette, diluted with deionized water to the measurement, and agitated. Lastly, the cuvettes were subjected to a 15-minute colorimetry study in the spectrophotometer.

Immunofluorescence assay

We employed immunofluorescence to ascertain the expression and localization of certain proteins in PC12 cells. When growth on the cover slip reached 95–100 %, the cells were taken out of the incubator. Following three 10-minute rounds of washing with 1x PBS, the cells were permeabilized for two to five minutes using 0.2 % Triton X-100. Three times, the cells were washed for ten minutes each using one PBS. The cells were fixed for 20–30 min at room temperature using 4 % formaldehyde after being cleaned three times for ten minutes each using

pre-warmed 1x PBS. Following three 10-minute rounds of washing with 1x PBS, the cells were permeabilized for two to five minutes using 0.2 % Triton X-100. Lastly, the cells were cultured for 30 min at room temperature with 5 % bovine serum albumin (BSA). For 4 min, add the primary antibody that has been diluted with 1 % BSA to a wet chamber. Subsequently, add the secondary antibody, diluted with 1 % BSA, and let it sit in the dark for half an hour. Wash three times for ten minutes each with 1 x PBS. Rewash three times for ten minutes each using 1 x PBS. Use 95 % glycerol to block. Please take note that 1 x PBS was used to dilute 4 % formaldehyde, 0.2 % Triton, and 5 % BSA for this experiment. Immunofluorescent antibodies name: Ubiquitin (P4D1) Mouse Antibody (CST, #3936, dilution 1:1000), GAPDH (D16H11) XP® Rabbit Antibody (CST, #5174, dilution 1:5000), Anti-FACL4 antibody (Abcam, ab227256, 1:200), and Goat Anti-Rabbit IgG H&L (Abcam, Alexa Fluor® 594) preadsorbed (Abcam, ab150084, 1:1000).

Western blot (WB) assay

We used several WB assays to measure the expression levels of ferritin heavy chain 1 (FTH1), glutathione peroxidase 4 (GPX4), and ACSL4. Protein extracts were isolated via SDS-PAGE gel electrophoresis and then deposited onto PVDF membranes from either brain tissue or cell cultures. The membranes were then blocked and incubated for an entire night with primary antibodies that were directed against the following proteins: GAPDH (D16H11) XP® Rabbit Antibody (#5174, dilution 1:5000) was acquired from Cell Signaling Technology. Anti-ACSL4 (ab155282, dilution 1:1000), anti-GPX4 (ab125066, dilution 1:2000), and anti-FTH1 (ab183781, dilution 1:1000) were purchased from Abcam (USA). The next day, we used an enhanced chemiluminescence kit to determine the amounts of protein expression and added secondary HRP-conjugated antibodies. By comparing groups, we were able to quantify the intensity of each protein band using image analysis tools and ascertain the impact of TQHX on ferroptosis marker proteins.

Flow cytometry assay

In PC12 cells grown under OGD/R conditions, flow cytometry was utilized to measure several cellular processes such as ferroptosis, apoptosis, and cell cycle analysis. Re-suspend cells at a concentration of $1\text{--}5 \times 10^6$ cells/ml in ice-cold PBS with 10 % FCS and 1 % sodium azide. To reduce cell loss during supernatant removal, stain cells in 12×75 mm² polystyrene round-bottomed Falcon centrifuge tubes and centrifuge completely. Put 0.1–10 $\mu\text{g/ml}$ of fluorescence-coupled primary antibody in there. If necessary, dilute the antibody with 3 % BSA/PBS. Keep the inoculum dark and at room temperature or 4°C for a minimum of half an hour. After centrifuging the cells for five minutes at 400 G, resuspend them in 500 μl to 1 ml of ice-cold PBS containing 1 % sodium azide and 10 % FCS. Repeat the procedure three times. Cells should be kept chilled or at 4°C with light protection until the analysis day.

Cell transfection

We validated that TQHX inhibited tMCAO injury by ACSL4 degradation while overexpressing ACSL4 reversed the treatment of TQHX in OGD/R-induced PC12 cells injury. The design of the sh-ACSL4 and OE-ACSL4 vector were templated with the cDNA sequence of the rat ACSL4 gene, following the mRNA design principles. PC12 cells were seeded in 6-well plates (2×10^5 per well) and allowed to adhere for 24 h before transfection. The cells were transfected with OE-ACSL4 while the sh-ACSL4 vector was injected into rat brain. Eventually, transfection efficiency expressing ACSL4-mRNA was assessed by flow cytometry.

Co-Immunoprecipitation (Co-IP)

After the cells were lysed, the quantity of proteins was assessed using

an immunoprecipitation buffer including inhibitors of phosphatase and protease. For one hour at 4°C, the cell lysates were pre-clared using protein A/G agarose beads. After incubating Protein A/G agarose beads for two hours at 4°C, the supernatant was subjected to either standard IgG as a negative control or an anti-ACSL4 antibody. The beads were boiled with the rising buffer subsequent to being cleaned with IP buffer. Using an ACSL4-specific antibody, protein blotting was carried out, and non-specifically bound proteins were then extracted. After that, the leftover protein complexes were separated from the beads and examined. Using antibodies against ACSL4, a WB analysis was performed to examine the protein complexes in more detail.

Quantitative real-time PCR (RT-qPCR) analysis

Using a Trizol reagent, total RNA was isolated from the colorectal tissues. Thereafter, a spectrophotometer was applied to assess RNA quality. Reverse transcription was performed on the isolated RNA using a PrimeScript™ RT reagent Kit (Perfect Real Time) (Takara; RR037Q). The obtained cDNA was stored in a refrigerator at -80°C. To carry out the amplification reaction, TB Green® Premix Ex Taq™ II FAST qPCR was applied (Takara; CN830S). Table S2 provides a summary of the primer sequences used in the qPCR.

Statistical analysis

The findings were presented in the form of mean ± standard deviation (SD). One-way ANOVA and two-way ANOVA with a significance level of $p < 0.05$ were carried out on various groups using GraphPad Prism version 8 software (GraphPad Software, CA, USA), followed by a post hoc Tukey's multiple comparison test.

Results

Quantitative analysis of TQHX extract

The composition of TQHX was shown in Fig. 1. Sorted in ascending

order of retention time were six typical TQHX compounds with the validated HPLC method: Hydroxysafflor Yellow, Oxypaeoniflorin, Paeoniflorin, Chlorogenic Acid, Ferulic Acid, and 1,2,3,4,6-O-Pentagalloylglucose (Fig. 1A and B). The content of these compounds was respectively determined to be 0.09 %, 0.07 %, 0.12 %, 0.03 %, 0.04 % and 0.10 %.

CIRI in rats: neuroprotective effects of TQHX

According to the dosage of TQHX, animals were randomized into five groups, then acclimatized for seven days and subjected to tMCAO modeling. To verify the effects of TQHX on cerebral I/R injury, neurological scoring tests were formally performed after seven days of continuous gavage administration, after which brain tissues were collected (Fig. 2A).

Firstly, based on Longa scores, the tMCAO group exhibited notable neurological abnormalities compared to the Sham group (Fig. 2B, $P < 0.01$). Next, to assess the extent of cerebral infarction, TTC staining was performed on the brain slices. The results showed that no infarct area was detected in the Sham group, whereas the tMCAO group exhibited a markedly increased converted infarct volume. However, the TQHX-H group demonstrated a remarkable reduction in the cerebral infarct volume relative to the tMCAO group (Fig. 2C). In detail, an increase of 30 % in infarct volume was displayed on the tMCAO group. In comparison, the TQHX-M and TQHX-H groups both showed a significant reduction to less than 20 % in infarct volume (Fig. 2D, $P < 0.01$). Moreover, TQHX-H treatment group revealed a significantly smaller infarct area than TQHX-M treatment group. Then, the amount of water in the brain was detected while a clear difference between the tMCAO and Sham groups was observed. Besides, TQHX-H group showed a tangible decrease in brain water content, achieving the same effect as Edaravone group, while TQHX-L group and TQHX-M group showed no significant changes (Fig. 2E, $P < 0.01$). Following that, Evans blue extravasation assay was employed to examine the integrity of the blood-brain barrier. According to the data, the blood-brain barrier disruption was notably exacerbated in tMCAO group, which could be reversed remarkably after

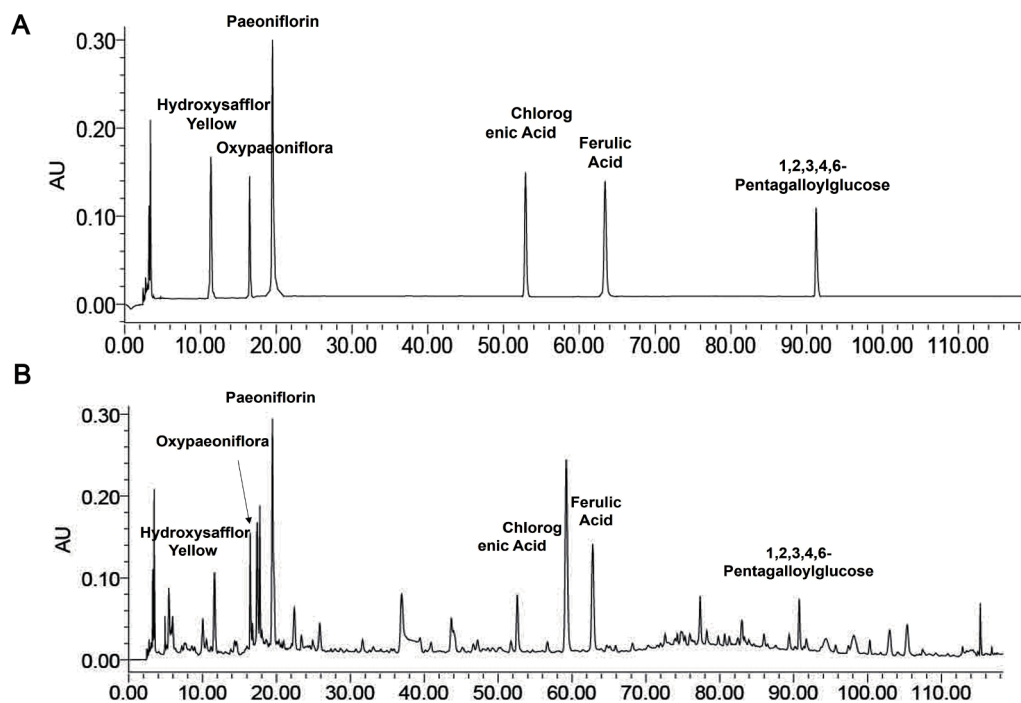


Fig. 1. TQHX extracts' representative HPLC chromatograms. Representative HPLC chromatograms of TQHX extracts are shown, featuring 6 key components: Hydroxysafflor Yellow, Oxypaeoniflorin, Paeoniflorin, Chlorogenic Acid, Ferulic Acid, and 1,2,3,4,6-O-Pentagalloylglucose. (A) standard compounds. (B) TQHX extract.

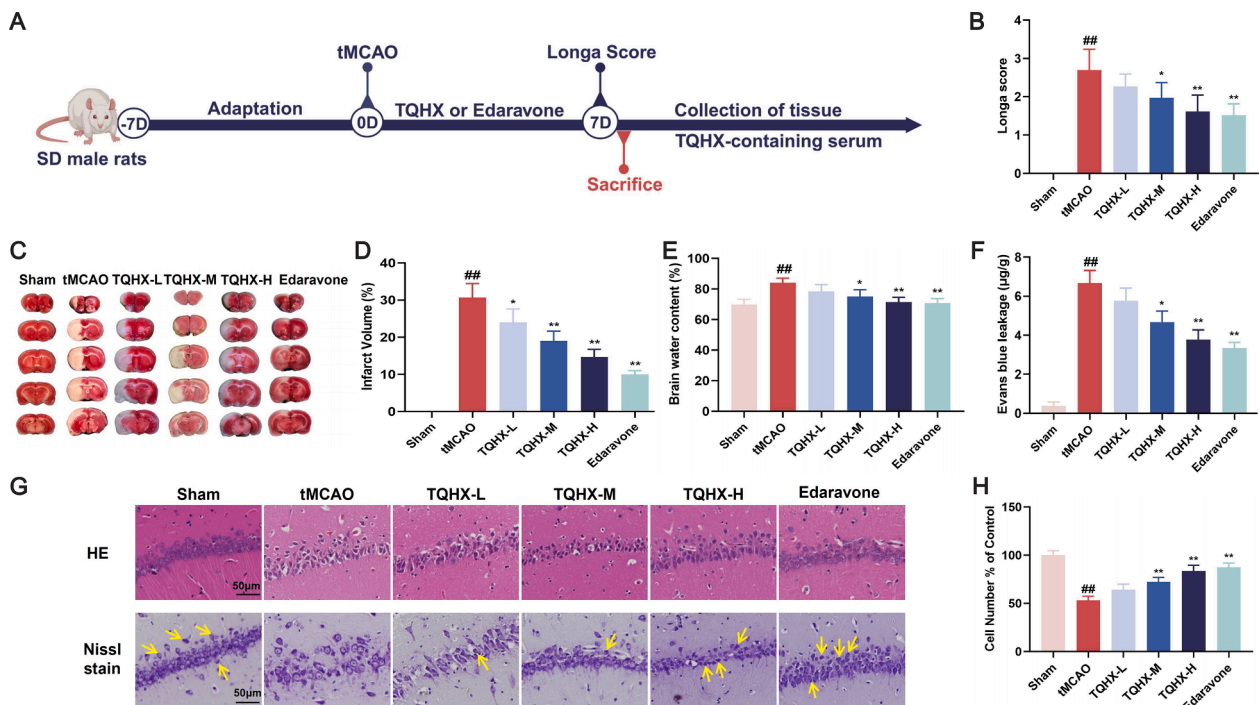


Fig. 2. TQHX's protective effects on cerebral infarction volume and neurological deficit after tMCAO in rats. (A) The pattern diagram illustrated how the tMCAO model was established and how pharmacological treatment was designed. (B) Neurological deficit score analysis (Longa Score). (C) The volume of cerebral infarction in rats by TTC Staining. (D) Detection of cerebral infarct volume. (E) Brain water content in rats. (F) Detection of brain tissue leakage with Evans Blue leakage. (G) Representative images by HE Staining and Nissl Staining of brain tissue. (H) The proportion of normal neuronal cells detected by Nissl Staining. Scale bar: 50 μ m. Results are presented as means $\pm$ SD. $n=6$, one-way ANOVA, $^{##}p < 0.01$, vs Sham group; $^{*}p < 0.05$, $^{**}p < 0.01$, vs tMCAO group.

TQHX treatment (Fig. 2F, $P < 0.01$). By applying further histological techniques of Hematoxylin and Eosin (HE) staining and Nissl staining to the rat brain tissue, as shown by the results, the number of intact neurons significantly decreased in the tMCAO group, whereas it was

positively correlated with the doses of TQHX (Fig. 2G). At last, the ratio of cell number in the ischemic semi-dark band was counted, the outcome of which revealed nearly no reduction in Sham group, while the ratio decreased to about 50 % in tMCAO group, and increased materially in

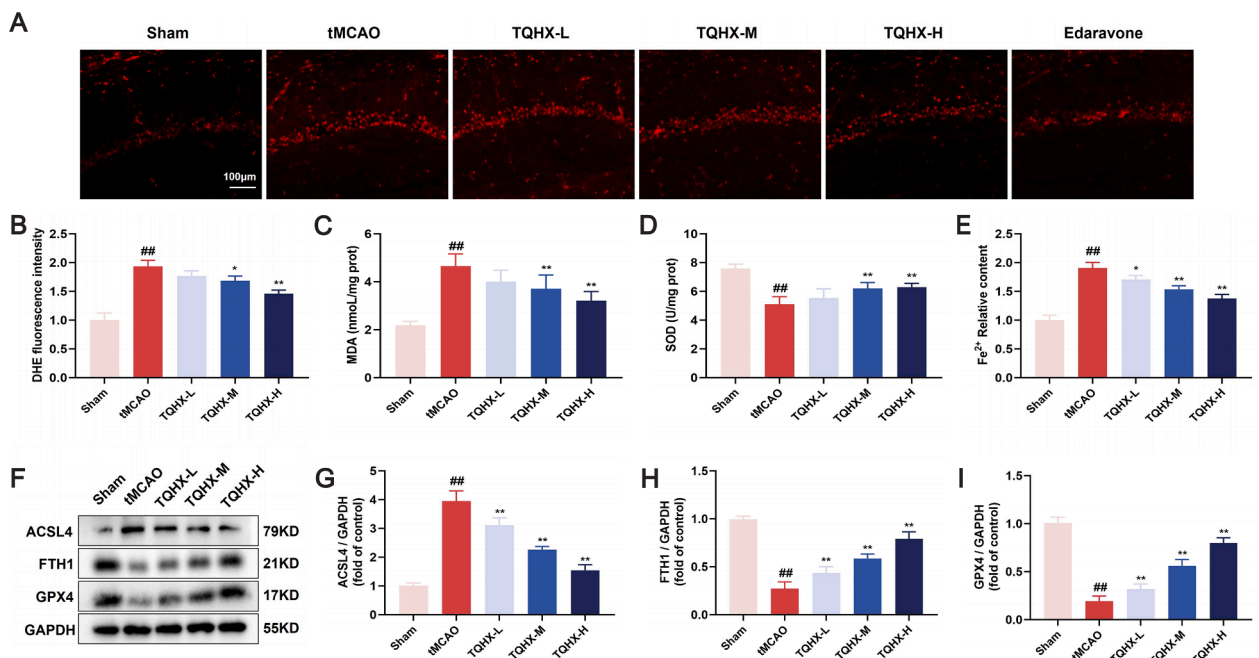


Fig. 3. In CIRI rats, TQHX inhibits CIRI-induced oxidative stress and ferroptosis. (A) ROS fluorescence staining of the hippocampal region of rat brain tissue. (B) ROS fluorescence quantification. (C) MDA content in the cortical area of rat brain tissue by Enzyme-Linked Immunosorbent Assay. (D) SOD content in rat brain tissue cortical area by Enzyme-Linked Immunosorbent Assay. (E) Fe²⁺ content in cortical areas of rat brain tissue by Colorimetric assay. (F) Expression of ferroptosis-related proteins in cortical areas of rat brain tissue by WB assay. (G-I) WB assay of ACSL4, FTH1, and GPX4 expression. Scale bar: 100 μ m. Results are presented as means $\pm$ SD. $n=6$, one-way ANOVA, $^{##}p < 0.01$, vs Sham group; $^{*}p < 0.05$, $^{**}p < 0.01$, vs tMCAO group.

TQHX-M group and TQHX-H group, up to 89.73 %, close to the treatment effect of Edaravone group (Fig. 2H, $P < 0.01$).

Collectively, cerebral I/R injury encapsulated neurological impairments and neuronal necrosis, while these findings indicated that TQHX reduced the size of the cerebral infarction, improved neurological performance, raised the number of neurons, implying the protective

influence of TQHX against CIRI.

TQHX inhibits CIRI-induced oxidative stress and ferroptosis

To validate TQHX's suppression of oxidative stress, we monitored the ROS status in the hippocampal region of the brain tissue through

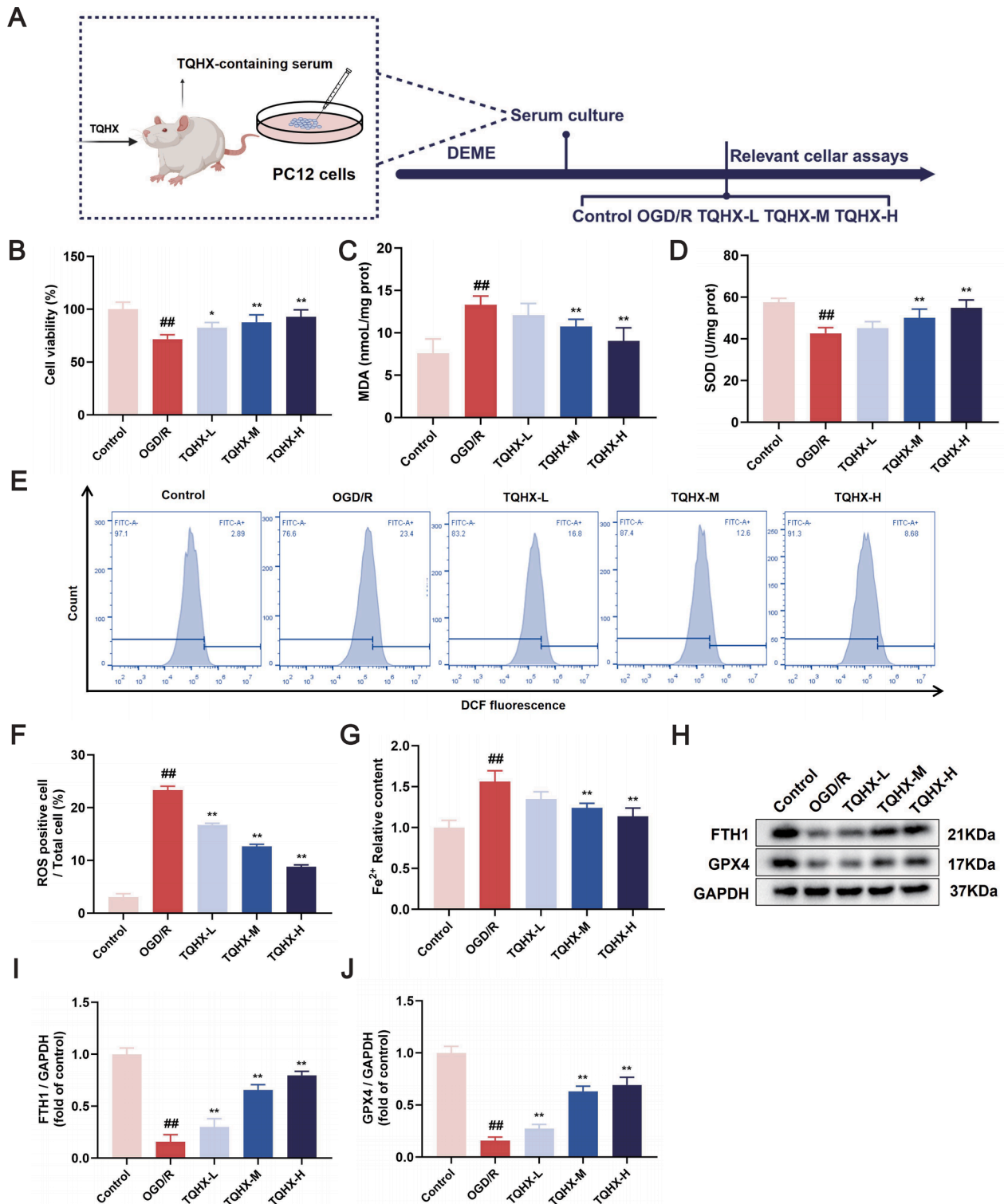


Fig. 4. TQHX protects OGD/R-induced PC12 cells injury by inhibiting ferroptosis. (A) Schematic diagram of TQHX feeder cells. (B) Detection of PC12 cell viability in 5 %, 10 %, and 15 % drug-containing serum by MTT assay. (C) MDA content in PC12 cells. (D) SOD content in PC12 cells. (E) Flow Cytometry of ROS in PC12 cells. (F) Proportion of ROS-positive cells to total cells. (G) Fe²⁺ relative content in ROS cells. (H-J) Expression of ferroptosis-related proteins in PC12 cells. Results are presented as means $\pm$ SD. One-way ANOVA, ## $p < 0.01$, vs Control group; * $p < 0.05$, ** $p < 0.01$, vs OGD/R group.

fluorescent probes, shown in Fig. 3A. First, we monitored the ROS status in the hippocampal region of the brain tissue (Fig. 3A). After CIRC, the amount of ROS in tMCAO group increased markedly. However, in comparison to the tMCAO group, the red fluorescence density showed a noteworthy decrease in the TQHX treatment groups (Fig. 3B, $P < 0.01$). The result illustrated that the accumulation of ROS was effectively inhibited by TQHX. Subsequently, the content of MDA and SOD in the cortical area of rat brain tissue showed that TQHX-M group and TQHX-H group noticeably reduced MDA and significantly increased SOD compared with those before treatment (Fig. 3C and D, $P < 0.01$). In contrast to the tMCAO group, the application of TQHX caused a stepwise decline in the amount of Fe^{2+} in the brain with increasing TQHX doses, thus indicating that TQHX has an inhibitory effect on ferroptosis (Fig. 3E, $P < 0.01$). To further corroborate this point of view, we conducted a WB assay to analyze the levels of ferroptosis-related proteins ACSL4, FTH1, and GPX4 expression in the cortical area of rat brain tissue (Fig. 3F). It was found that the expression of both FTH1 and GPX4 (Fig. 3H and I, $P < 0.01$) in the TQHX administration group increased with increasing dose. Nevertheless, the level of ACSL4 exhibited a significant reduction (Fig. 3G, $P < 0.01$) compared with the tMCAO group.

Accordingly, we reasonably speculated that TQHX served as an inhibitor of CIRC-induced oxidative stress and ferroptosis in CIRC injury.

TQHX exerts a protective effect on PC12 cells by inhibiting ferroptosis

Following ferroptosis in CIRC rats, Fe^{2+} accumulation leads to ROS and inflammatory response, which in turn triggers cell damage and death. To further confirm TQHX's impact by hindering ferroptosis in rats, we made a dependable OGD/R experimental model that simulates

the CIRC process in vivo (Fig. 4A) and kept an eye on a number of pertinent indications, such as cell viability, the amount of MDA and Fe^{2+} , and the relative ROS level. According to the amount of low to high drug in the serum, TQHX treatment groups were divided into three, including the TQHX-L group (5 %), TQHX-M group (10 %), TQHX-H group (15 %). Subsequently, the cell viability of PC12 in 5 %, 10 %, and 15 % drug-containing serum was detected by MTT assay, and it revealed that the cell viability was dramatically lower in the OGD/R group and increased with increasing concentrations of TQHX (Fig. 4B, $P < 0.01$). The amount of MDA and SOD in PC12 cells was then measured using ELISA. The outcomes showed that the TQHX therapy significantly lowered the levels of MDA, with the TQHX-H group showing better effectiveness than the OGD/R group (Fig. 4C, $P < 0.01$). On the contrary, the level of SOD exhibited a noteworthy increase upon drug treatment (Fig. 4D, $P < 0.01$). Moreover, Flow Cytometry was performed for ROS in PC12 cells in each group (Fig. 4E), indicating the ratio of ROS-positive cells revealed significantly improvement in the OGD/R group, which showed remarkable reduction following TQHX treatment (Fig. 4F, $P < 0.01$). TQHX administration successfully and dose-dependently restored the Fe^{2+} elevation that was seen in the OGD/R group (Fig. 4G, $P < 0.01$). Meanwhile, WB assay was applied for the expression of ferroptosis-related proteins FTH1 and GPX4 in PC12 cells (Fig. 4H). The expression of FTH1 was demonstrated to significantly suppress in the OGD/R group while noticeably activated in the three groups after treatment with TQHX, so as GPX4 (Fig. 4I and J, $P < 0.01$). To make the results more credible, we researched further at the transcription level concerning oxidative stress and ferroptosis. According to Fig. 5A, TQHX-L or TQHX-M did not influence significantly on the ACSL4 mRNA expression. However, the mRNA expressions of FTH1, GPX4 and SOD1

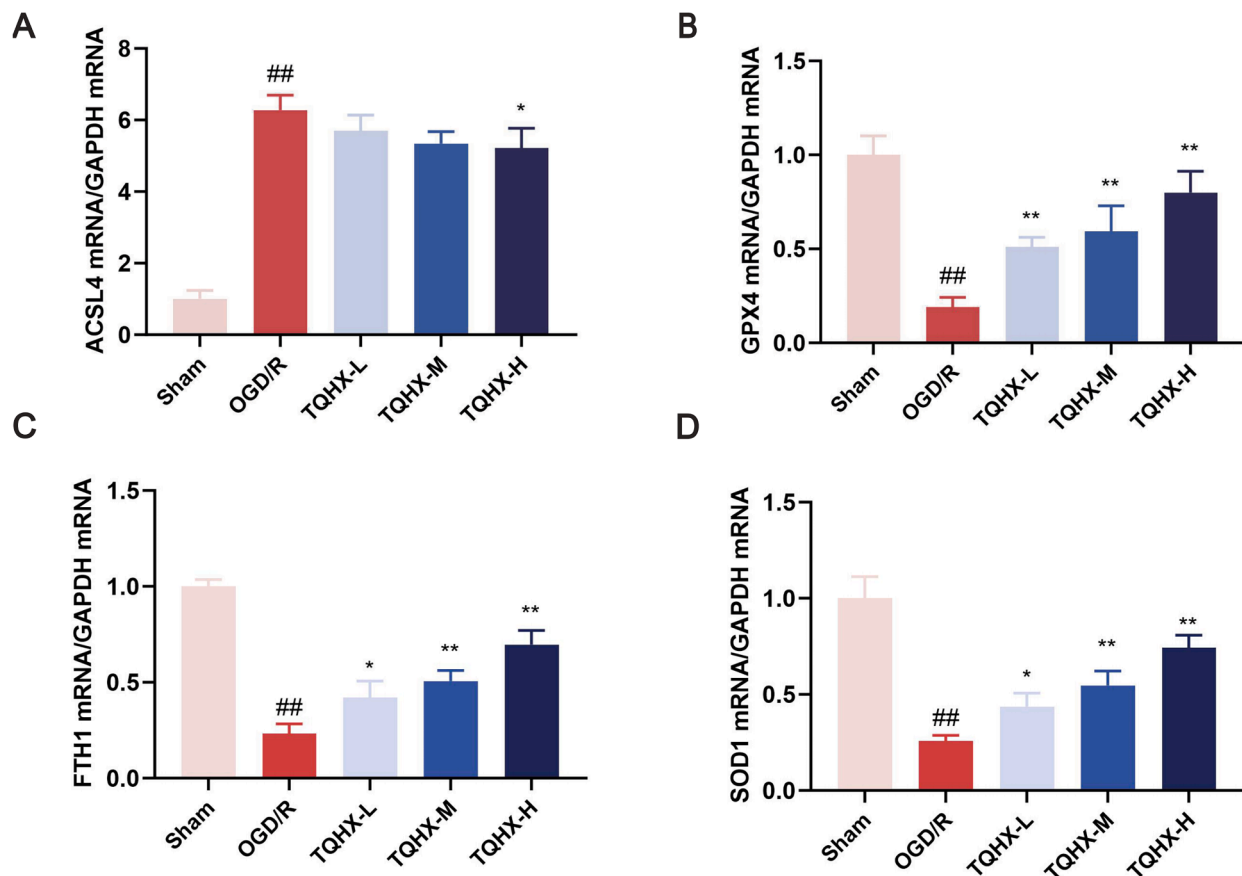


Fig. 5. The mRNA expression of ferroptosis-related proteins in PC12 cells. (A) The mRNA level of ACSL4 in PC12 cells. (B) The mRNA level of GPX4 in PC12 cells. (C) The mRNA level of FTH1 in PC12 cells. (D) The mRNA level of SOD1 in PC12 cells. Results are presented as means $\pm$ SD. One-way ANOVA, ## $p < 0.01$, vs Sham group; * $p < 0.05$, ** $p < 0.01$, vs OGD/R group.

gradually raised with increasing doses of TQHX (Fig. 5B-D, $P < 0.01$).

In conclusion, the analysis in vitro above reconfirmed the ability of TQHX in inhibiting oxidative stress and ferroptosis in a stoichiometry-dependent manner.

TQHX inhibits CIRI-induced ferroptosis by promoting the degradation of ACSL4 protein

ACSL4 is a crucial regulator of ferroptosis, to further confirm that the degradation of ACSL4 protein could prevent ferroptosis in cells, we conducted Immunofluorescence staining to measure the expression and localization of ACSL4 in PC12 cells (Fig. 6A). Our analysis revealed a significant increase in the protein fluorescence intensity of ACSL4 in the OGD/R group. Notably, compared to the OGD/R group, that showed a substantial decrease with escalating doses of TQHX (Fig. 6B, $P < 0.01$). The expression level of ACSL4 in the cells was then further examined (Fig. 6C), and it was found that ACSL4 protein band strips in the TQHX treatment group had a weaker fluorescence intensity than those in the OGD/R group, according to the WB assay. The expression of ACSL4 was considerably reduced in the TQHX-H group (Fig. 6D, $P < 0.01$). The downregulation of the level of ACSL4 lessened the harm that ferroptosis

does to the brain's tissue.

Meanwhile, to further investigate the in-depth mechanism of TQHX which affects ACSL4 to ultimately inhibit ferroptosis, we conducted in vitro assays in which both groups (the Model + CHX group and Model + CHX + TQHX group) were treated with the protein synthesis inhibitor CHX (2 $\mu\text{g}/\text{ml}$). Analysis of the protein bands corresponding to the two groups displayed enormous decline over time, representing successful in vitro modeling (Fig. 6E). When the two groups were compared, it was found that the expression of ACSL4 in the Model + CHX + TQHX group with the addition of TQHX intervention was materially lower than that in the Model + CHX group without TQHX intervention at 0.5 and 1 hour of testing. It was noteworthy that this difference was alleviated at 2 h of testing (Fig. 6F, $P < 0.01$).

The above experiments provided evidence that TQHX inhibited PC12 cells from undergoing ferroptosis by promoting ACSL4 protein degradation, which may be related to the inhibition of ACSL4 protein synthesis.

TQHX promotes the ubiquitination and degradation of ACSL4 in OGD/R

To further explore how TQHX affects the degradation of ACSL4

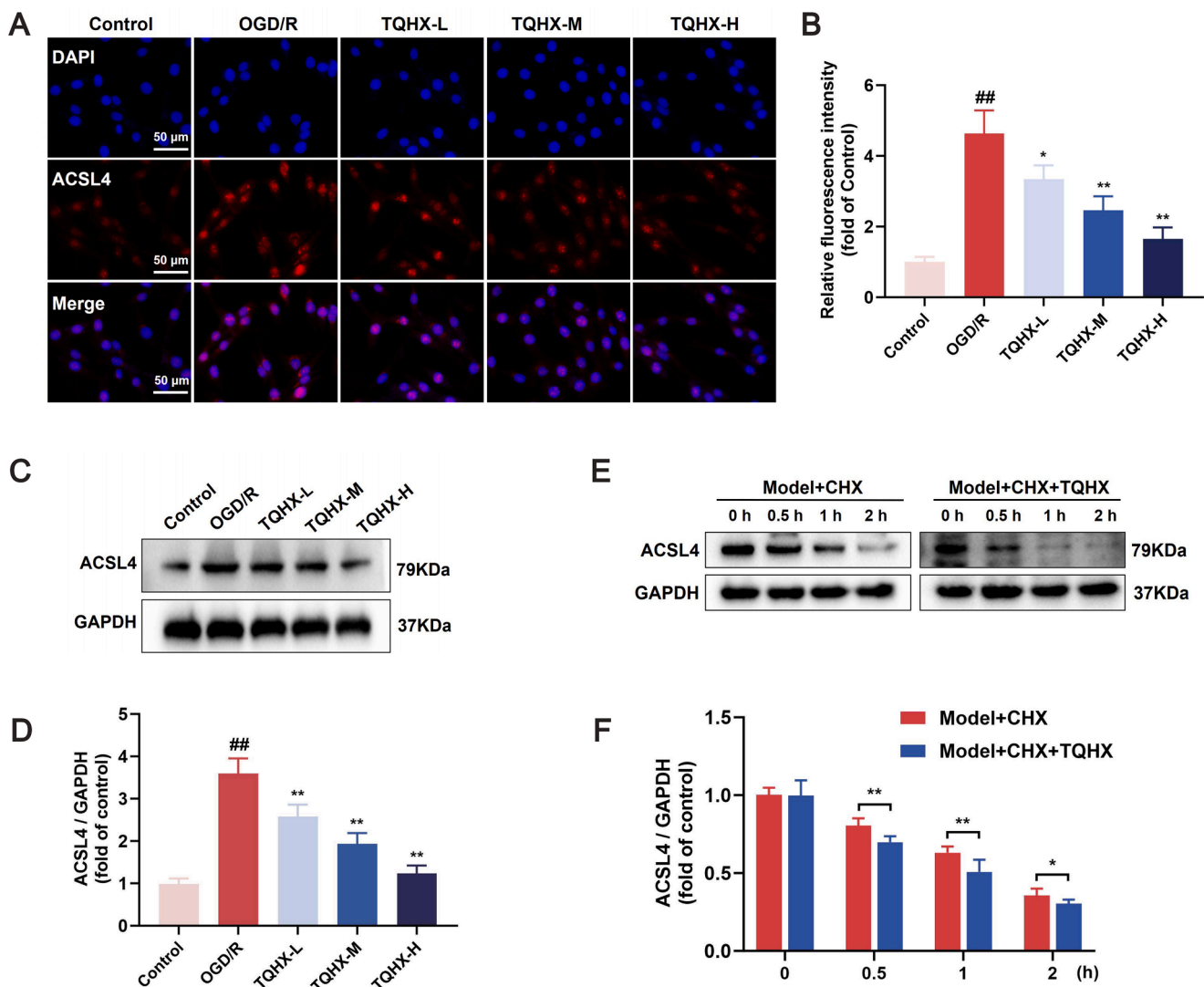


Fig. 6. TQHX promotes the degradation of the ACSL4 protein and inhibits CIRI-induced ferroptosis. (A-B) Relative fluorescence density of ACSL4 in cells by Immunofluorescence assay. Representative images are shown. C Representative WB bands of ACSL4. D Relative protein expression of ACSL4. (E-F) In vitro experiments to verify the mechanism of ACSL4 degradation. Scale bar: 50 μm . Results are presented as means $\pm$ SD. One-way ANOVA, ^{##} $p < 0.01$, vs Control group; ^{*} $p < 0.05$, ^{**} $p < 0.01$, vs OGD/R group.

protein, we then added the proteasome inhibitor MG-132 (25 $\mu\text{g}/\text{ml}$) for observations. TQHX demonstrated neuroprotective properties in a dose-dependent manner according to our earlier studies, thereby a high dosage of TQHX was employed here to uncover the importance of the findings. WB assay was operated to examine the effect of MG-132 on ACSL4 degradation in OGD/R (Fig. 7A). When opposed to the group without TQHX-H and MG-132 intervention, intracellular ACSL4 expression was noticeably suppressed in the group that was in the presence of TQHX-H only, without MG-132 intervention (Fig. 7B, $P < 0.01$). Nonetheless, there was no noteworthy difference when the two groups both included MG-132 but one was TQHX-H embodied while the other was not (Fig. 7B, $P < 0.05$). Additionally, the ubiquitin level of ACSL4 was detected through Co-IP (Fig. 7C). The administration of escalating doses resulted in a steady elevation in Ub/ACSL4 expression levels, with the TQHX-H group demonstrating the greatest levels of expression in comparison to the control group (Fig. 7D, $P < 0.01$).

It could be concluded plausibly that the mechanism by which TQHX inhibited ACSL4 protein content might be related to the process of intracellular protein synthesis rather than the inhibition of related proteases. Meanwhile, ubiquitination degradation may play a major role.

Overexpression of ACSL4 reverses the effect of TQHX on ferroptosis in OGD/R

Based on the experiments above, it has been determined that ACSL4 played a significant role in inhibiting oxidative stress and ferroptosis caused by CIRI. Subsequently, we presumed that overexpression of ACSL4 may obtain the opposite outcome. The overexpressed ACSL4 group had much higher ACSL4 expression, as indicated by the OGD/R cell model through the WB assay (Fig. 8A), implying successful ACSL4

overexpression (Fig. 8B, $P < 0.01$). According to the results evaluated by MTT assay, the cell viability of PC12 in the OGD/R + TQHX-H group was significantly raised because of TQHX's neuroprotective impact, compared with the OGD/R group. Besides, the OGD/R + OE ACSL4 + TQHX-H group displayed a remarkable decrease in cell viability, reversing the result of TQHX (Fig. 8C, $P < 0.01$). Next, we found that the ratio of ROS-positive cells noticeably increased in the OGD/R group by detecting ROS in the PC12 cells of each group. Moreover, the OGD/R + OE ACSL4 group had higher levels of basic ROS in PC12 cells. However, the OGD/R + OE ACSL4 + TQHX-H group and OGD/R group had lower ratio of positive cells than the OGD/R + OE ACSL4 group did. The OGD/R + TQHX-H group owned the most ROS positive cells except of the control group (Fig. 8D, $P < 0.01$), which was detected by the Flow Cytometry (Fig. 8E). Then, we assessed the Fe^{2+} content changes in PC12 cells by ELISA assay. Similarly, the OGD/R group exhibited a substantial increase in Fe^{2+} , whereas the OGD/R + TQHX-H group showed a significant opposite result. At the same time, the OGD/R + OE ACSL4 + TQHX-H group displayed a remarkable increase tendency in Fe^{2+} compared with the OGD/R + TQHX-H group (Fig. 8F, $P < 0.01$), indicating that ACSL4 overexpression suppressed TQHX's inhibition effect on ferroptosis. Finally, TQHX considerably enhanced the protein levels of ferroptosis-related proteins FTH1 and GPX4 in PC12 cells (Fig. 8G), according to a WB study. In comparison with the OGD/R + TQHX-H group, not only the expression in the OGD/R + OE ACSL4 group substantially reduced, but also in the OGD/R + OE ACSL4 + TQHX-H group (Fig. 8H and I, $P < 0.01$), revealing that overexpressed ACSL4 reversed the expression level of FTH1 and GPX4. Increased lipid peroxidation caused by ACSL4 overexpression can cause neuronal ferroptosis by reducing the expression of FTH1 and GPX4.

In a word, overexpression of ACSL4 reverses the effect of TQHX on ferroptosis in OGD/R.

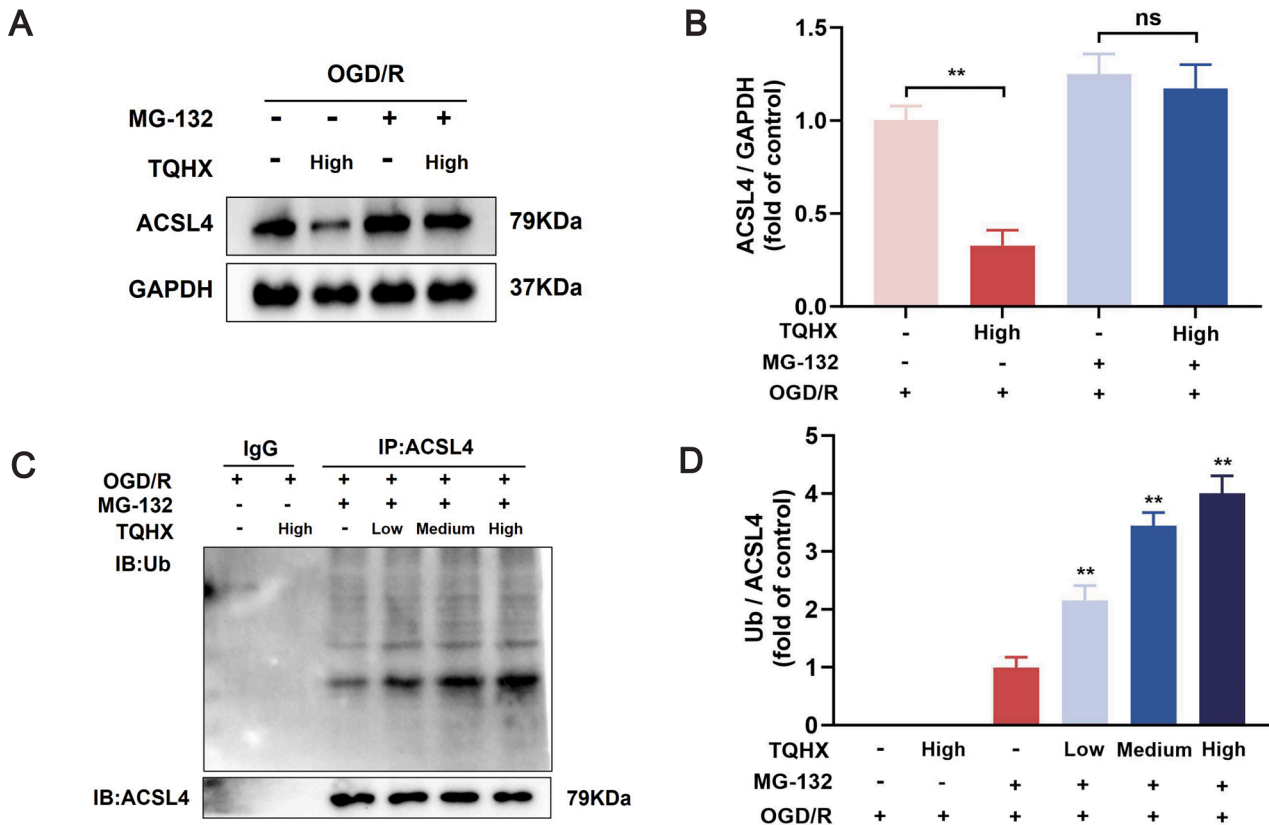


Fig. 7. OGD/R model shows that TQHX promotes ACSL4 ubiquitination and degradation. (A-B) WB assay for the effect of MG-132 on ACSL4. (C-D) Ubiquitination status of ACSL4 protein in cells by Co-IP. Results are presented as means $\pm$ SD. Two-way ANOVA, *** $p < 0.01$, vs Control group; ns, no significant difference.

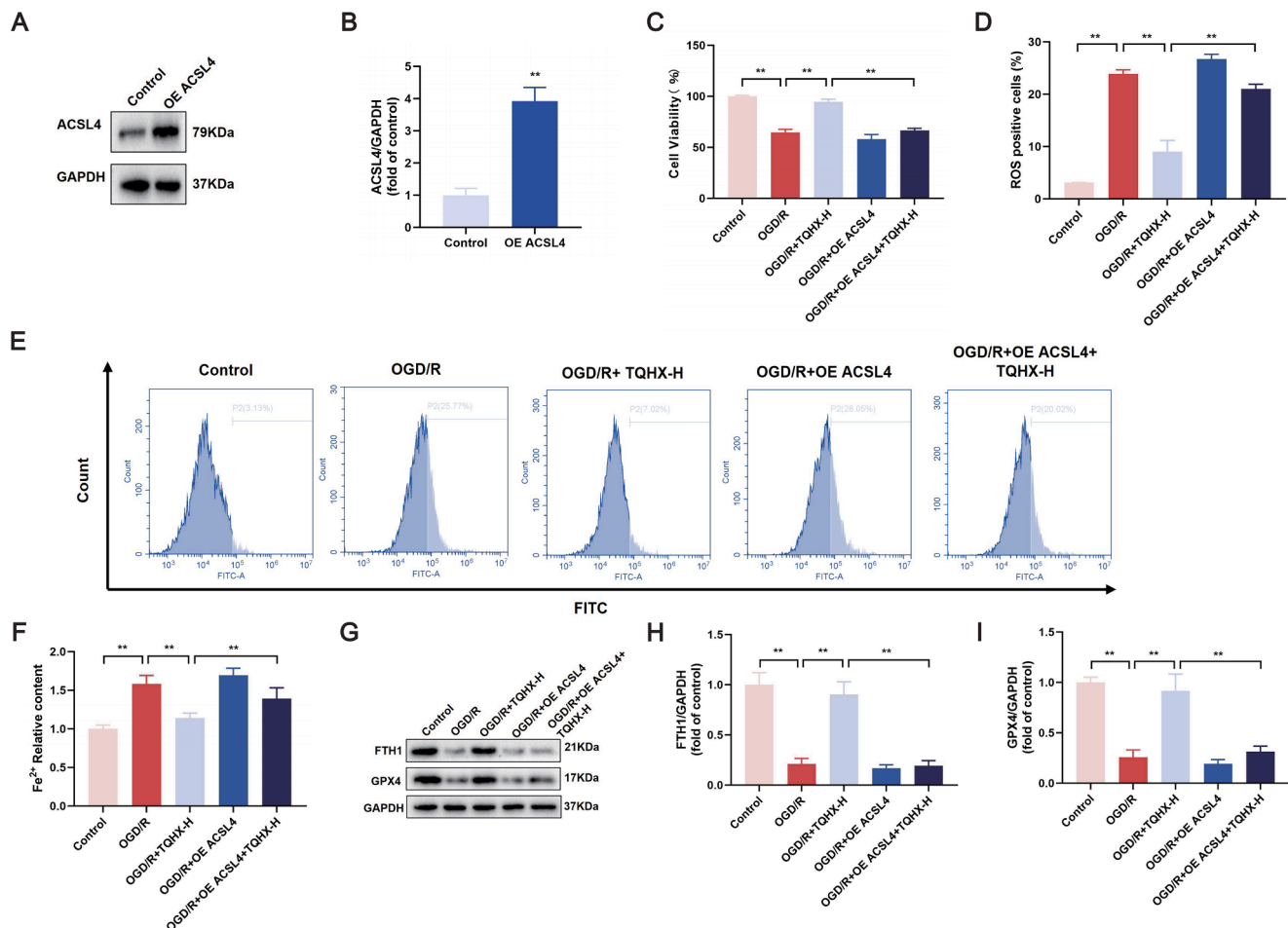


Fig. 8. Overexpression of ACSL4 reverses TQHX's regulation on ferroptosis. (A-B) Detection of OE ACSL4 in OGD/R model by WB assay. (C) Detection of PC12 cell viability by MTT assay. (D) Proportion of ROS-positive cells to total cells. (E) Flow cytometry assay of ROS in PC12 cells. (F) Fe^{2+} content in ROS cells. (G-I) Expression of ferroptosis-related proteins in PC12 cells by WB assay. Results are presented as means $\pm$ SD. One-way ANOVA, * $p < 0.05$, ** $p < 0.01$, vs Control or OGD/R group.

Silence of ACSL4 owns neuroprotective effects and abolishes CIRI-induced oxidative stress and ferroptosis

ShRNA-ACSL4 vectors were constructed to silence the ACSL4 gene in the hippocampal region of the brain tissue, aiming at reconfirming the connection between ACSL4 and TQHX treatment on CIRI. The model was established smoothly based on Fig 9A-B, in which the expression of ACSL4 in the shACSL4 group had a significant decline (Fig. 9B, $P < 0.01$). Subsequently, a noteworthy diminution in the infarct volume was observed in the shACSL4 group in comparison with the tMCAO group (Fig. 9C). It is remarkable that no substantial difference was found between the shACSL4 group and the shACSL4 + TQHX-H group (Fig. 9D, $P < 0.01$). The similar results could be noticed when brain water content was detected (Fig. 9E, $P < 0.05$). Moreover, it was easy to observe notably enlarged and more plentiful Nissl bodies in the shACSL4 group (Fig. 9G) while the number of surviving neurons was obviously higher (Fig. 9H, $P < 0.01$). Compared with the shACSL4 group, the shACSL4 + TQHX-H group did not exhibit improved neurons count (Fig. 9H). The analysis above explained that ACSL4 silence owned neuroprotective effects and abolished CIRI-induced oxidative stress, meanwhile, the treatment effect TQHX took was closely related to ACSL4. To verify whether inhibition of ACSL4 influenced ferroptosis in CIRI or not, Fe^{2+} relative content and ROS fluorescence intensity were tested in succession (Fig. 9F and I-J). It turned out that the application of TQHX-H led to the remarkable decline in brain Fe^{2+} amount, meanwhile, the red fluorescence intensity suggested a considerable drop in the shACSL4 group in

comparison to the tMCAO group (Fig. 9F and J, $P < 0.01$). In accordance with expectation, when silenced-ACSL4 and TQHX treatment were simultaneous, no notable change could be surveyed between the shACSL4 and shACSL4 + TQHX-H group.

To be concluded, silence of ACSL4 abolish CIRI-induced oxidative stress and ferroptosis, reconfirming the inseparable relationship between ACSL4 and TQHX in the treatment of CIRI.

In summary, the above data indicated that ACSL4 promoted ferroptosis. Additionally, it appeared that TQHX could effectively inhibit ferroptosis and CIRI occurrence. Taken together, we concluded that ACSL4 was a key factor and TQHX could facilitate its ubiquitination degradation to mediate the regulation of ferroptosis.

Discussion

There is a growing body of evidence linking cerebral ischemia and neuroinflammation within the CIRI. In cerebral ischemia, disruption of blood supply causes cellular hypoxia and insufficient energy supply (Yang et al., 2023). Additionally, the neuroinflammatory cascade response causes apoptosis, disruption of the blood-brain barrier, cerebral edema, and hemorrhagic transformation, which exacerbate CIRI reactions (Ma et al., 2022).

For a long time, TCM prescriptions have been used to treat neurological disorders (Zhao et al., 2023b). Studies have shown that Buyang Huanwu decoction can improve CIRI through the regulation of Aprt and Pde1b (Yin et al., 2023), Longxuetongluo Capsule has been found to

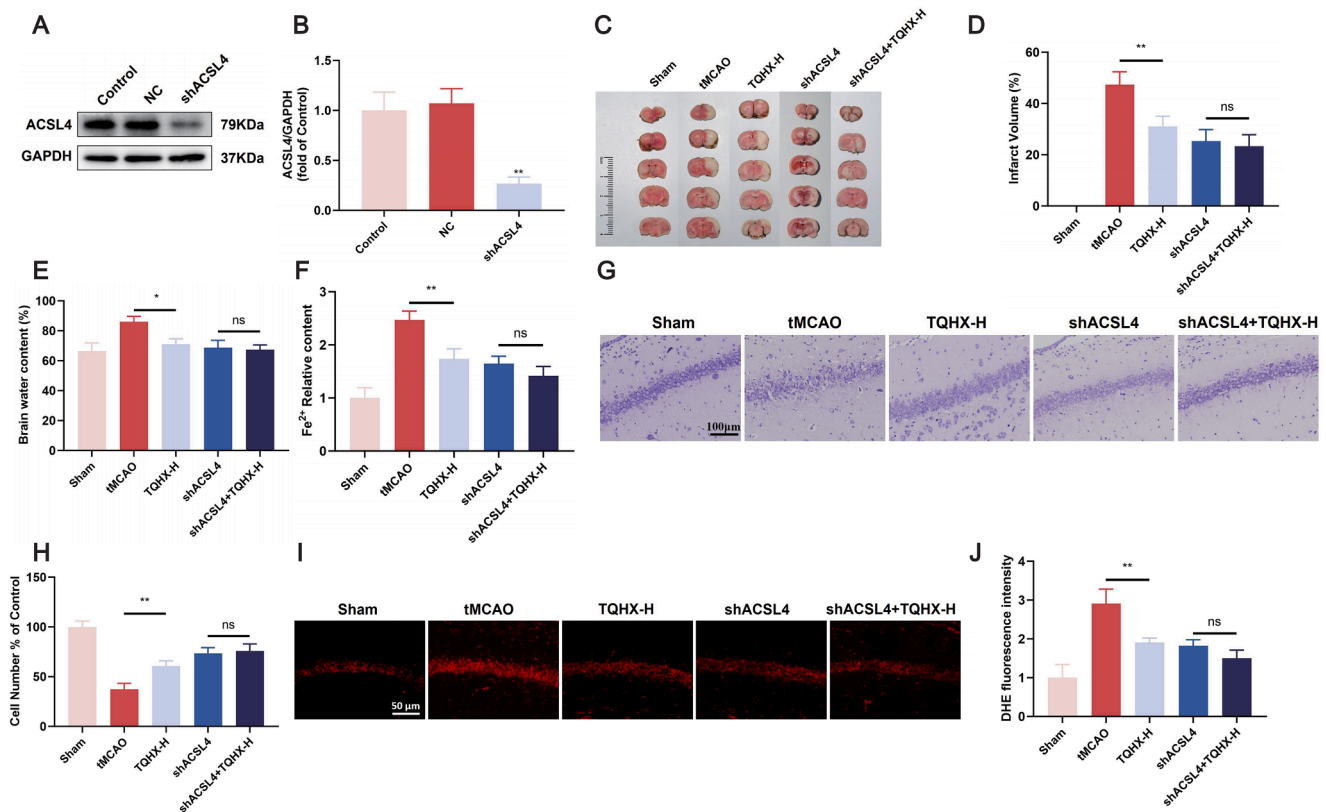


Fig. 9. Effects of silenced-ACSL4 on neurological deficit and ferroptosis after tMCAO in rats. (A–B) Detection of sh ACSL4 in tMCAO model by WB assay. (C) The volume of cerebral infarction in rats by TTC Staining. (D) Detection of cerebral infarct volume. (E) Brain water content in rats. (F) Fe²⁺ content in cortical areas of rat brain tissue by Colorimetric assay. (G) Representative photographs of Nissl-stained brain sections. Scale bar, 100 μ m. (H) Analysis of neurons number. (I) Representative fluorescent images of ROS. Scale bar, 50 μ m. (J) Mean fluorescence intensity of ROS. Results are presented as means $\pm$ SD. $n = 6$, one-way ANOVA, * $p < 0.05$, ** $p < 0.01$, vs tMCAO group; ns, no significant difference, vs shACSL4 group.

prevent CIRI-induced damage by utilizing MAPK-mediated pathways and endoplasmic reticulum stress (Pan et al., 2021) and Zhilong Huoxue Tongyu Capsule has also been shown to improve CIRI through the regulation of the AGE-RAGE signaling pathway (Li et al., 2022). In conclusion, TCM prescriptions emerge as effective interventions for addressing CIRI.

In our investigation, tMCAO and OGD/R were applied as in vivo and in vitro models of brain CIRI based on prior studies to examine the neuroprotective benefits of TQHX and its potential mechanism. The findings offered convincing evidence that TQHX may prevent brain CIRI damage by reducing oxidative stress and ferroptosis in a dose-dependent way, hence preserving the blood-brain barrier's integrity and lowering the death of apoptotic cells.

Previous studies have revealed that following a cerebral ischemic stroke, oxidative stress was a major factor causing brain tissue damage (Zhang et al., 2023a). Since brain tissues contain a lot of iron, unsaturated fatty acids, and oxygen, they are vulnerable to oxidative damage. Massive levels of ROS are produced by neuronal cells during ischemia and the following reperfusion, which overwhelms their removal and upsets the antioxidant systems' equilibrium (Hong et al., 2023). When oxidant generation in CIRI exceeds the body's antioxidant capacity, the resulting build-up of ROS will initiate the oxidation process of DNA, lipid, and protein, ultimately resulting in tissue cell death and neurological impairment (Zhang et al., 2023a). As a consequence, inhibition of ROS accumulation and prevention of the occurrence of oxidative stress may confer neuroprotective effect, providing an important therapeutic strategy. Moreover, oxidative stress manifests as decreased levels of antioxidant defense enzymes like CAT, SOD, and GPX and higher indicators like MDA or lipid hydroperoxides (Chen et al., 2022). In our research, intracellular ROS, SOD, MDA levels were measured in

vivo and in vitro. The results demonstrated that TQHX had the ability to reverse CIRI injury-induced and OGD/R-induced ROS and other oxidative stress markers increase.

Furthermore, it has been established that ferroptosis is closely connected with oxidative stress (Chen et al., 2022). This particular mode of cell death, which is caused by iron-dependent phospholipid peroxidation, is controlled by a variety of cellular metabolic processes, such as iron handling, redox homeostasis, mitochondrial function (Jiang et al., 2021). Recently, ferroptosis is becoming more widely recognized as a process behind cell death during ischemia in brain, impacting neuronal injury during CIRI (Tuo et al., 2022).

Arachidonic acids (AAs), which are among polyunsaturated fatty acids (PUFAs), are the primary substrates of lipid peroxidation in the process of ferroptosis, which can result in neuronal injury (Zheng and Conrad, 2020). However, on the background of the loss of the ACSL4, unesterified arachidonate (AA) improves. Ferroptosis cannot be induced by such peroxidation, even though free AA is still susceptible to it (Liang et al., 2022). Additionally, GPX4 is an antioxidant enzyme that functions as a downstream mediator of amino acid antiporter system xc⁻. It reduces phospholipid hydroperoxides in membranes to provide cytoprotection against ferroptosis (Xue et al., 2023). Previous studies have demonstrated that overexpression of FTH1 hampered ferritin autophagy and suppressed nuclear receptor coactivator 4 and microtubule-associated protein light chain 3 expression, which in turn prevented ferroptosis-induced cell death (Tian et al., 2020). Lipid peroxidation and iron buildup are two of ferroptosis's main biological characteristics. In our investigation, we discovered that TQHX attenuated the oxidative stress response by dramatically regulating the expression of important ferroptosis-related proteins above.

Further investigations into the mechanism of ACSL4 reduction

determine that the main cause is due to ACSL4 ubiquitination, thereby modulating ferroptosis, presenting a viable therapeutic target for CIRI (Ji et al., 2024). Primarily, ubiquitination level of ACSL4 in PC12 cells was measured. Meanwhile, in order to certify efficacy of ACSL4 in TQHX treatment of ferroptosis, we innovatively made overexpression ACSL4 in PC12 cell and silenced ACSL4 in rat brain. Our findings ultimately proved that overexpressing ACSL4 reversed the therapeutic effect of TQHX in OGD/R-induced PC12 cells. However, inhibited-ACSL4 promoted neurological function and suppressed ferroptosis in tMCAO neurons, providing significant evidence that TQHX abolishes ferroptosis in nerve cells by facilitating ACSL4 ubiquitination degradation.

Nevertheless, this study has several limitations. The text confirms the overall effectiveness of TQHX on CIRI but does not provide evidence for the activity of TQHX's components. There are still many obstacles to overcome in the advancement of TQHX, including the lack of further research on its systemic active components. Additional comprehensive research is required to perform more animal behavioral tests, to include more samples in each group, and additional review relevant clinical reports in conjunction with or in contrast to clinical medications.

Conclusions

To be concluded, we discovered that in the animal model, TQHX lessened blood-brain barrier breakdown, decreased infarct volume, and improved neurological impairments. Moreover, whether in vivo or in vitro, TQHX efficiently reduces oxidative stress, intracellular ROS production, and ferroptosis-related protein expression. What's important, we elucidated a novel molecular mechanism whereby TQHX facilitates ACSL4 protein degradation through ubiquitination. These results provide a promising therapeutic pathway on ischemic stroke for TQHX, via targeting the ACSL4 protein.

CRediT authorship contribution statement

Zhijie Ou: Visualization, Validation. **Yanting Deng:** Writing – original draft, Writing – review & editing. **Yan Wu:** Formal analysis. **Yuanqi Wang:** Methodology, Investigation. **Yijing Zhao:** Data curation. **Chang Liu:** Resources, Project administration. **Zhuoyu Wang:** Supervision, Software. **Manhua Liu:** Visualization, Validation. **Xin Hu:** Formal analysis. **Li Fang:** Investigation. **Juping Chen:** Data curation, Conceptualization.

Declaration of competing interest

There are no conflicts of interest to declare.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2024.155701](https://doi.org/10.1016/j.phymed.2024.155701).

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